PQC DAILY STUDY SERIES

# PQC Adoption Gap in 2026: Browser Readiness vs. Server-Side Lag

**Day 12** | Category: PQC Adoption & Industry Status

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## PQC Adoption in 2026: The Gap Between Browser and Server-Side Readiness

In 2026, the transition to Post-Quantum Cryptography (PQC) has reached a critical and somewhat uneven phase. While major web browsers have become quantum-ready, the broader server-side infrastructure is lagging significantly. This creates a notable gap, where the full security benefits of PQC cannot be realized until both clients and servers are in sync. Today, we'll explore this adoption gap, analyze the metrics, and discuss its implications for internet security.

### Client-Side Success: Browsers Lead the Charge

The client-side of the PQC transition has been a remarkable success. All major web browsers now ship with hybrid post-quantum key exchange enabled by default. This proactive approach ensures that a significant portion of internet users have the capability to establish quantum-resistant connections.

> The primary hybrid mechanism in use is **X25519Kyber768**, which combines the classical Elliptic Curve Diffie-Hellman (ECDH) key exchange (X25519) with the NIST-selected PQC algorithm ML-KEM (formerly Kyber). This ensures that even if one algorithm is compromised, the connection remains secure.

### Server-Side Lag: The Root of the Adoption Gap

While browsers are ready, they can only establish a PQC-secured connection if the web server they are communicating with also supports the new hybrid cipher suites. This is where the adoption gap becomes apparent. The rollout of PQC on the server-side is far slower and more complex, influenced by factors like software update cycles, hardware limitations, and risk management priorities.

PQC Adoption Metrics (Mid-2026 Estimates)

| Domain | Browser Support (Client-Side) | Server Support (PQC Key Exchange) |
|---|---|---|
| Chrome, Firefox, Safari, Edge | ~100% (Default Enabled) | N/A |

| Domain | Browser Support (Client-Side) | Server Support (PQC Key Exchange) |
|---|---|---|
| Top 100 Websites | N/A | ~40% |
| Top 1 Million Websites | N/A | ~8.6% |
| Enterprise Internal Services | N/A | < 2% |

### Why Does This Gap Matter?

This discrepancy has several critical implications for cybersecurity:

- **Incomplete Protection:** A browser's PQC capability is dormant until it connects to a PQC-enabled server. This means that a large portion of web traffic remains vulnerable to "Harvest Now, Decrypt Later" (HNDL) attacks, where adversaries store encrypted data today to decrypt it with a future quantum computer.
- **Uneven Security Landscape:** The gap creates a two-tiered internet. Connections to major tech platforms (Google, Cloudflare, etc.) are increasingly quantum-resistant, while connections to smaller businesses, public sector services, and internal enterprise applications are often not.
- **Delayed Certificate Transition:** The focus so far has been on hybrid key exchange in TLS 1.3. The next major step, issuing publicly trusted PQC digital certificates (using ML-DSA/SLH-DSA), has not yet begun. This transition is expected to start with pilot programs in late 2026 and gain traction in 2027.

### Daily Quiz

1. What is the primary hybrid key exchange mechanism being adopted by major web browsers in 2026?
2. What does the term "Harvest Now, Decrypt Later" (HNDL) refer to, and why is it relevant to the PQC adoption gap?
3. According to 2026 estimates, what percentage of the top 1 million websites have enabled server-side support for PQC hybrid key exchange?